

Ethan Woo

647-918-6941 | ecpwoo@uwaterloo.ca | [LinkedIn](#) ↗ | [GitHub](#) ↗

TECHNICAL SKILLS

Languages: ARM-Assembly, C, C++, C#, Java, MATLAB, Python, SQL, VHDL

Libraries: Autoware, Eigen, Gazebo, Gym, Matplotlib, NumPy, OpenCV, Pandas, Pangolin, PyBullet, PyTorch

Technologies/OS: Arduino, AWS, Debian, Docker, GCP, OnShape, Solidworks, STM32, ROS2, RTOS

EXPERIENCE

University of Waterloo Control, Learning, and Logic Group

Sep 2024 – Present

Undergraduate Research Assistant

Waterloo, ON

- Developed self-driving Jeep Cherokee autonomous vehicle control system using **ROS2** and **Carla** simulators.
- Collaborated on the setup and interface of autonomous driving stacks using **Apollo** and **Autoware**.

Advanced Electrophoresis Solutions

May 2024 – Aug 2024

Embedded Systems Developer Intern

Cambridge, ON

- Implemented a specialized operating system on a custom circuit using industry standard tools (**PetaLinux**).
- Wrote device drivers using various protocols (**RS-232/485**, **I2C**, **SPI**) in **C**, improving system runtime by 15%.
- Reduced build/computation time by 20% through efficient process handling and memory management in **C++**.
- Programmed automation **Python** scripts to interface with medical devices, increasing testing speeds by 12%.

Epson Canada

Jan 2023 – Apr 2023, Aug 2023 – Dec 2023

Software Developer Intern

Markham, ON

- Developed algorithms to capture golf ball movements and calculate their spin and velocity (**OpenCV**, **C++**).
- Designed a golf simulator with 95% accurate readings while reducing costs by 80% compared to competitors.
- Enhanced data capturing and debugged on robotic arms for bin-picking and computer-vision ML models.

University of Waterloo Aerial Robotics Design Team

Sep 2022 – Jan 2024

Autonomy Robotics Developer

Waterloo, ON

- Led a 6-person team to build an interface and a piloting program for 3D vision on aerial drone.
- Contributed to AI recognition model with 90% accuracy using **PyTorch** on a Jetson Tx2.
- Utilized multithreading to improve drone communications while adhering to standard **PEP8** formatting.

PROJECTS

SLAM-Spacemapper ↗

Apr 2024 – Present

- Designed an autonomous robot using **OnShape** with an **ORB-SLAM** algorithm in **C++** with **OpenCV**.
- Interfaced a **Raspberry Pi** and Arducam to send feed via TCP, visualized map using **Pangolin**.

Quadrupedal Simulation ↗

Mar 2024 – Jul 2024

- Created a robot simulation in **PyBullet** to train a reinforcement learning **DDPG** network for four-leg walking.
- Utilized **ROS2** and **UART** to implement control code and the **RL algorithm** to the quadrupedal robot.

Movie Generator Transformer ↗

Jan 2024 – Mar 2024

- Built a decoder-only transformer architecture neural network to generate random movie plots using **PyTorch**.
- Implemented multi-headed self-attention and trained off Wikipedia movie summaries.

EDUCATION

University of Waterloo

Sep 2022 – Present

Bachelor of Applied Science Mechatronics Engineering - 3.92 GPA

Waterloo, ON

Amazon Web Services

Sep 2023 - Present

Amazon Certified Cloud Practitioner